



A Novel Hierarchical Pipeline for Fine-Grained Punch Recognition in Uncontrolled Setting

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Abstract. Human Action Recognition (HAR) is one of the most emerging topics in the field of Computer Vision. It finds utilization in a wide range of applications like athletics, healthcare, security, sports performance, etc. In sports, HAR can be used to monitor athletes' movements to facilitate skill development and decision-making training. However, in case of highly dynamic sports like boxing, automatic HAR becomes more difficult. Variations in lighting conditions, background, camera perspectives, and rapid movements collectively compound the complexity in automatic HAR and its performance further worsens with the involvement of uncontrolled environment videos. This manuscript proposes a hierarchical pipeline for robust sports activity recognition. For this purpose, a novel regression-based detection is introduced using a Dual Stream Spatio-Temporal Transformer, and a 3D Convolutional Neural Network (CNN) is used for fine-grained classification. The approach commences with regression-based detection to identify punch instances within unedited boxing video streams. Subsequently, a fine-grained classification module is employed to categorize detected punches into six basic punch types: jabs, cross, lead, and rear hooks, or lead and rear uppercuts. This hierarchical architecture enables efficient and accurate punch recognition in diverse real-world wild settings. Experimental results on various unedited YouTube videos show the effectiveness of our proposed approach. The proposed approach achieves an overall mean detection accuracy of 96.09% and a mean classification accuracy of 88.25% on test videos.

Keywords: Sports Activity Recognition · Regression · Classification · Spatio-temporal Transformer

1 Introduction

Human Activity Recognition (HAR) is one of the major research domains in Computer Vision. It finds application in various fields of research like healthcare, anomaly detection, sports analytics, human-computer interaction, etc. [2, 9]. HAR in sports makes it easier to analyze games tactically and physically and evaluate athletes' performances for the coaches and the sports world at large [12]. Despite this growing need, there is limited research on combat sports like boxing. In settings where sports are competitive, like boxing, the post-match analysis of the number of different kinds of punches attempted provides a valuable source of information.

Initially, statistics in various combat sports like boxing were generated by human annotators. It required hours of human efforts to minutely segment the punches for its analysis [14]. Various sporting arrangements also used Inertial Measurement Unit (IMU) based sensors, which consist of gyroscopes and accelerometers, for punch classification. IMU sensors come with various disadvantages, like wearing them during boxing training or matches may disrupt an athlete's focus or comfort and sometimes lead to various injuries [6].

With Vision-based approaches, actions can be detected automatically, thus reducing human intervention. But many visual-based approaches use depth cameras, which are costly and thus are not effective for historical videos [6, 24, 32]. Thus, there is a need for the development of an accurate punch classifier pipeline suitable for *in-the-wild* settings to understand the total number of punches a boxer attempts during a bout. This will also provide a base for the future development of punch analyzers. The following are the major contributions:

- A hierarchical-based approach for punch classification from unedited videos is proposed. The two-step approach involves detecting the fixed-length punch clips from the input video and, finally, classifying the detected punch clip into one of the six basic punch categories.
- Considered the task of punch detection as a regression problem rather than classification, which allows us to quantify the partial actions. Employing a sigmoid activation function in the final layer restricts the output probabilities within the interval $[0, 1]$, wherein a value of 1 denotes the inception of a punch.
- Demonstrated the utility of the proposed methodology on different unedited videos collected from YouTube. Distinct sets of train and test videos featuring different subjects were selected to underscore the generalizability of the proposed approach.

2 Related Work

Human motions can be represented as sequences of poses, encapsulating spatio-temporal data. Human Action Recognition (HAR) is a vibrant and rapidly evolving field with diverse applications, from enhancing human-computer interactions to bolstering security systems. The computational interpretation of such data

offers extensive benefits across diverse domains like surveillance [4] [13], Human-Computer Interaction [20], sports analytics [28] [17], and medicine [7, 33]. In this section, we delve into two pivotal approaches: vision-based and sensor-based methods, each representing a substantial body of research and innovation.

Several studies [3, 14, 27, 30] have showcased the utility of Human Action Recognition (HAR) in individual and team sports, facilitating player training and fair assessment. Even with simple MLP models, accurate kinematic data can lead to decent action recognition accuracy regardless of the players' proficiency levels [14]. In recent years, the advancements in deep learning techniques within computer vision have notably enhanced video-based sports HAR [30], providing satisfactory outcomes applicable to various individual and team sports, thereby augmenting player training and evaluation protocols. Deep learning models surpass traditional machine learning approaches [3] due to their capacity to capture intricate data patterns through extensive parameterization. However, the sensitivity of deep learning models to dataset characteristics, especially with small dataset sizes, remains a challenge.

2.1 Sensor-Based HAR

In sensor-based HAR, sensors capturing the kinematics of specific body parts during actions are utilized. For example, in boxing, Inertial Measurement Unit (IMU) sensors placed at key body locations such as wrists, elbows, and shoulders provide precise 3D kinematic data, including linear and angular velocities and accelerations. This results in multiple time series data, which can be classified using various techniques such as Convolutional Neural Networks (CNNs) [11], Long Short-Term Memory networks (LSTMs) [21], statistical embeddings like Multiview Cauchy Estimator Feature Embedding (MCEFE) [8], and Physics-aware adversarial networks [23], among other deep learning approaches [16, 29], on accelerometer data for action classification.

Although sensor-based methods achieve satisfactory accuracy, obtaining on-body sensor data during combat sports presents challenges due to potential damage during bouts. Furthermore, collecting data during official matches is often impractical due to restrictions on using add-ons to equipment during matches, making on-body sensor-based evaluation during matches unfeasible.

2.2 Vision-Based HAR

Visual data is categorized into two domains based on their dimensionality, i.e., 2D and 3D (with depth) images. 2D imagery is straightforward to capture using any low-end camera system. However, they exhibit an intrinsic self-occlusion problem, resulting in the corrupted or missing pose estimation. 3D imagery is captured with multiple camera systems or Time-of-Flight (ToF) based cameras. It can capture RGB videos and depth information during motion despite occlusions. However, sophisticated and expensive vision sensors are required to obtain 3D images or videos that include ground truth depth information.

Vision-based human action recognition involves analyzing sequences of images and capturing subjects in various activities. This approach categorizes based on how images are processed:

RGB Image-Based Action Recognition. Several studies, such as Mohith et al. [22] and Abdul et al. [1], harnessed trajectory descriptors from optical flow to classify general human actions. However, these approaches demanded substantial computational resources, limiting their real-time applicability. On the other hand, Yi et al. [32] adopted dense-trajectory descriptors, correlation, and weighted trajectory encoding as features extracted from 2D visual data, achieving accurate human action classification. Stefański et al. [26] labeled a 50 FPS video of an Olympic match, employing clash detection to remove frames with significant player distance. Their CNN model classified actions into the head, corpus, and block categories, lacking fine-grained boxing punch classification and sequential context.

Skeleton-Based Action Recognition. Skeleton-based Human Action Recognition (HAR) has emerged as a more promising approach, focusing on human landmarks as perceptible cues for action understanding. Luo et al. [18] introduced a time-series modeling of HAR by integrating Pose Estimation with Long Short-Term Memory (LSTM) networks. Silva et al. [24] proposed a rapid human action recognition method using visual question-answering techniques, processing a 2D image frame in a mere 190 milliseconds. Their approach involved an ensemble of Bayesian, Multinomial, Complement, and Gaussian classifiers. These classifiers were applied to same-side and cross-side Euclidean distances between selected joints and knee, elbow, and shoulder joint angles. Additionally, relative keypoint coordinates concerning the nose, assumed to be a static point during the action, were incorporated as features. Yang et al. [31] employed graph convolutional neural networks during boxing actions, addressing frame coarseness with Neville interpolation. Conversely, Hudovernik et al. [10] used Vision Transformers in Jiu-jitsu videos, applying rule-based algorithms for automated scoring but facing challenges with covariate shifts across different fighting styles. Researchers like Shujah et al. [25] have also explored various human pose kinematic features for pose information encoding. One of the most active research topics in Artificial intelligence and Computer Vision is Mixed Reality, which allows humans to interact with real-world and computer-generated environments. It automates fitness and sports training in virtual, augmented, or mixed reality. Makarov and Petrov [19] developed a prototype of a Shadow Boxing simulator for automated control and gamification processes.

Recent works predominantly focus on general and weak human action classification with existing large datasets using proprietary controlled datasets. For fine-grained, highly dynamic action recognition, such as sports activity, highly accurate and 3-dimensional data is required, which can be achieved with the help of sensors. However, sensor-based methods are not applicable in real-time gameplay. Since vision-based are non-invasive, they can be used for sports activ-

ity recognition. Fine-grained action recognition requires accurate 3D poses, but it requires complex and expensive video acquisition systems. Classifying fine-grained sports action using 2D pose sequences is intrinsically challenging due to high self-occlusion and monocular view. To the best of our knowledge, sensor-based and vision-based methods employed controlled environment datasets captured with constrained settings, which are not generalized for real-time or historical videos. In-the-wild videos with a monocular view make the task of fine-grained analytics extremely formidable. We aim to design a multi-stage deep learning-based pipeline to tackle this problem of fine-grained sports activity recognition on monocular view videos in the wild setting. For this task, various types of boxing punch recognition is taken as a showcase example.

3 Methodology

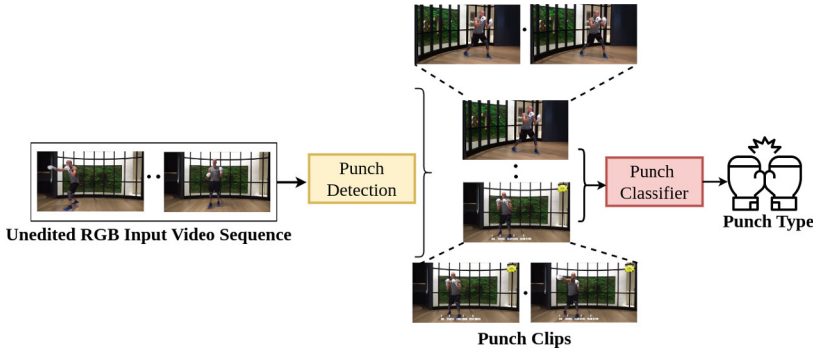


Fig. 1. The overall pipeline of the proposed methodology. The unedited input video is passed through the detection model, which returns fixed-length punch clips. The 2D poses of the punch clips are passed through the classification model for final classification.

3.1 Overview

The end-to-end pipeline for fine-grained boxing punch recognition is shown in Fig. 1. It takes an unedited RGB video of variable duration as input. Eventually, it provides the starting and ending frame numbers of detected punches and the punch class. The proposed pipeline is a deep learning-based hierarchical approach consisting of a deep learning-based detection module followed by a classification module. Firstly, the input video sequence is partitioned into multiple clips of fixed frame length and fed to the detection module. The module detects clips belonging to the punching activity, and the remaining clips are discarded. Punching activity is highly dynamic, varies from person to person, and depends on fps

and view angle. Therefore, direct binary classification of the clips into a punch and no punch would result in misleading and imprecise punch detection with respect to the frame number. To resolve this problem of accurate and precise detection of high-speed activity, i.e., punching in our case, we encouraged the problem statement as a regression task instead of classification. In this regression task, the probability of exhibiting a punch in the clip is computed using a deep-learning model. For model training purposes, all the clips are assigned with punch probability as target values instead of binary labeling. Here, we introduce the concept of two types of matching in action recognition problems, i.e., hard matching and soft matching. In hard matching, a pose sequence can have only two probability values as targets: 0 or 1. Meanwhile, in soft matching, a pose sequence is assigned with a target value equal to the probability of finding a particular action. This concept of soft matching enables partial action recognition and helps generalize feature representation learning. During the training process, our task is to learn the probability of having a punch within the input pose sequence. Learning confidence scores as probability empowers discriminative latent space representation to distinguish between a complete start-to-end punch and no punch adequately.

All these selected candidate clips are processed through a classifier module to find the specific or fine-grained type of punches. The proposed algorithm considers six basic punch categories: Jab, Cross, Lead Hook, Lead Uppercut, Rear Hook, and Rear Uppercut.

3.2 Network Architecture

Detector. As shown in Fig. 2, the Punch Detector module takes fixed-length (F frames) clips in RGB format as input and provides the frame numbers of detected punches as output. It first extracts 2D poses in each frame using AlphaPose [5]. AlphaPose outputs 17 key body joints in COCO. Since punching depends on the body’s extremities, 12 joints out of 17 joints are selected for the feature extraction stage. These critical 12 joints include shoulders, elbows, wrists, hips, knees, and ankles. Hereafter, 2D pose sequences are fed to a deep learning-based punch probability estimator. The backbone of the punch probability estimator is the Dual Stream Spatio-Temporal Transformer (DSTformer) [34], which allows motion encoding to capture dynamic features among skeleton keypoints. Initially, it encodes spatial and temporal features separately in a latent space and combines them to get spatiotemporal feature representation using the adaptive feature fusion technique. These features are converted into k -dimensional embeddings using a Fully Connected (FC) layer. An MLP-based regression head followed by a sigmoid layer is attached to the backbone to find the punch probability. To restrict probability values from 0 to 1, the sigmoid function layer is used to get the final output. To supervise this regression task of estimating the probability, Huber loss is employed, a combination of Mean Square Error (MSE) and Mean Absolute Error (MAE) 1.

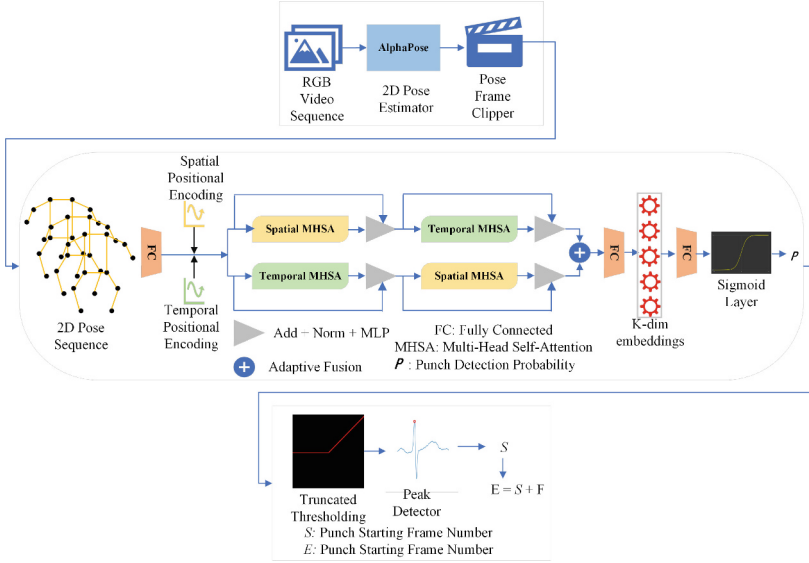


Fig. 2. The end-to-end Punch Detection module. The module is divided into three sub-modules: The upper one is the Pose Estimator, the Middle one is the Probability Estimator, and the lower one is the Peak Detector. Out of them, the probability estimator sub-module (middle) is trainable.

$$L_{\delta}(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2 & \text{if } |y - f(x)| \leq \delta \\ \delta(|y - f(x)| - \frac{1}{2}\delta) & \text{otherwise} \end{cases} \quad (1)$$

All the clipped pose sequences of an input video are fed to the punch probability estimator in time sequential format to obtain the probabilities as time series data. This output time series probability data undergoes truncated threshold operation to remove local maxima. The thresholded probability score as time series data is smoothened using a Savitzky-Golay filter with a window size 7 and a polynomial order of 4. The filter's input-output relationship equation is 2, where x_{k+i} represents the data points within the window centered at x_k and c_i are the coefficients of the polynomials that best fit the data points using the least squares method 3. Afterward, a gradient-based zero-crossing detector is applied to find the peaks or maxima throughout time series data. These peaks correspond to the clipped pose sequences, with the highest probability of detecting punches throughout the video.

$$y_k = \sum_{i=-3}^3 c_i \cdot x_{k+i} \quad (2)$$

$$\mathbf{C} = (\mathbf{X}^T \cdot \mathbf{X})^{-1} \cdot \mathbf{X}^T \cdot \mathbf{Y} \quad (3)$$

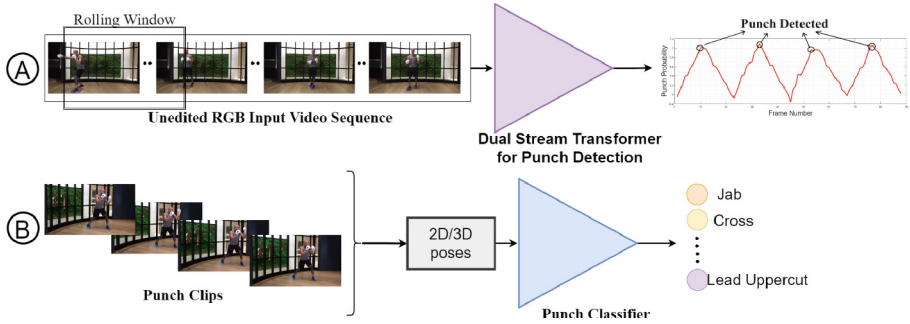


Fig. 3. (A) Punch detection algorithm (B) Classification algorithm.

Classifier. The classifier model is attached after the detector to classify the punches detected into six basic punch categories: Jab, Cross, Lead Uppercut, Rear Uppercut, Lead Hook, and Rear Hook, as shown in Fig. 3. A 3D Convolutional Neural Network (3D-CNN) is used as a classification module. As the previous subsection explains, a 3D-CNN is strategically selected to capture the spatiotemporal features from the 2D or 3D punch sequences detected using the detector. The 2D poses from the punch sequences are extracted using Alphapose, whereas the 3D poses are extracted using the MotionBert model. The classification module has three convolutional layers, each using kernels of size $3 \times 3 \times 3$. The first layer consists of 32 kernels, and in subsequent layers, the number of kernels is increased by a factor of 2 to learn hierarchical representations of input data. The early layers typically learn low-level features, while deeper layers learn more abstract and complex features [15]. By increasing the number of kernels in deeper layers, the classifier can capture more diverse and abstract features, allowing it to learn richer representations of the poses. ReLU activation function follows each convolutional layer. The final dense layer has a sigmoid activation function to squash the output values to the range $[0, 1]$, making it interpretable as probabilities.

4 Experiments

4.1 Dataset Description

The proposed pipeline takes an unedited RGB video of variable duration as input. Eventually, it provides the starting and ending frame numbers of detected punches and the punch class. As mentioned in the methodology section, the hierarchical-based approach detects the punch clips followed by their fine-grained boxing punch type classification. All the unedited videos considered for training and testing are collected from YouTube and are publicly available thus, no ethical clearance is taken for our experiments. The detector and the classifier are trained on <https://www.youtube.com/watch?v=KidSVNv0WcY&t=212s> (Subject 1, Video 1). The video (S11) is 15 min and 40 s long and consists of almost 415

punches. Subsequently, four videos of the same and different subjects are selected for further experiments to see how the model adapts to unknown subjects. The video collected from <https://www.youtube.com/watch?v=rIk9fylC0Rk> is of the same subject named S12 (Subject 1, video 2). The video <https://www.youtube.com/watch?v=wB8GhqtKLdk&list=LL&index=3&t=11s> is the second video from the test set and is named S21 (Subject 2, video 1). Similarly, <https://www.youtube.com/watch?v=o9qPILLGv6k&list=LL&index=4&t=156s> is named S31 and <https://www.youtube.com/watch?v=aXaKD-jSiDQ&t=257s> is named S41. Further, the dataset preparation for the detector and classifier is explained in more detail below.

Detector. The video stream is manually annotated for training purposes with exact starting and ending frame numbers of punches, irrespective of their types. Afterward, the video is partitioned into multiple clips of definite length frames ($F = 11$) with a rolling window of size 1. Let's assume there are N punches annotated from the training video, $L = P_1, P_2, \dots, P_{N-1}, P_N$, where P_i represents the set of frames from starting to ending of the punch i . C_t is the clip partitioned at time instant t . Each clip is assigned with a confidence score P_{C_t} governed by the Eq. 4.

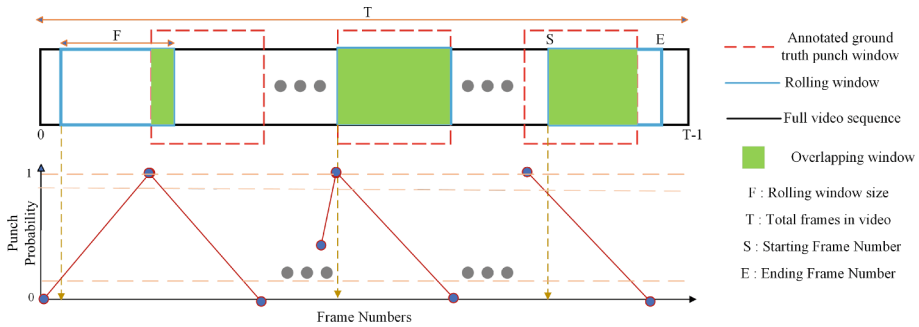


Fig. 4. Labelling formulation of the training dataset to compute probability score of each video clip

$$P_{C_t} = \frac{\sum_{i=1}^N |C_t \cap P_i|}{F} \quad (4)$$

Figure 4 shows that red-colored windows represent manually annotated starting and ending frame numbers, whereas colored windows correspond to the fixed-length clips. Clips are marked at the starting frame number, and the corresponding confidence values are arranged in the series graph below. There are a total 21305 number of clipped video sequences labeled with confidence scores for the training.

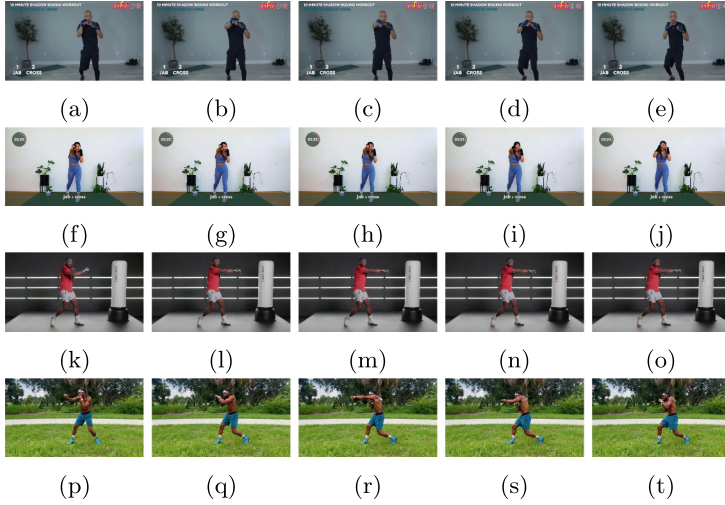


Fig. 5. (a)-(e) S11 (f)-(j) S21 (k)-(o) S31 (p)-(t), and S41 showing *Cross*.

Classifier. The video denoted as S11 was considered for training. Subsequently, the punches identified by the detector in the first stage were subject to manual categorization into six basic punch types, thereby constituting the training dataset. The ground truth was decided based on what the boxing trainer in the video said. However, in S11, various punch categories like lead hook, rear hook, lead uppercut, and rear uppercut were underrepresented. So, a few clips belonging to the underrepresented class from S12 were also considered for training. This is done to have a balanced dataset so that the final classifier is not biased towards a particular class. S21, S31, and S41 are considered for testing. All subjects' visuals are shown in Fig. 5. Notably, since all video materials used for our experiments consist of boxing training sessions, thus the ground truth annotations were derived from the spoken directives provided by the respective trainers.

4.2 Training Details

Detector. The detector module's deep learning model has a spatio-temporal transformer backbone. This backbone and the regression head are end-to-end trained on a single subject video. The spatiotemporal backbone is trained with depth $N = 5$, number of heads = 8, and feature and embedding dimension = 128. Input spatial and temporal feature dimensions are $J = 12$ and $T = 11$, respectively. The Adam optimizer is employed for training with a batch size 4028 for 300 epochs. The initial learning rate is 0.0001 with an exponential decay factor of 0.99 per epoch.

Classifier. The 3D (CNN) architecture for classification tasks consists of 595,110 trainable parameters. In the case of AlphaPose, the model’s input dimensions are defined as $11 \times 12 \times 2$, where 11 signifies the number of frames within the input clips, while 12 denotes the 2D pose coordinates derived from the AlphaPose framework. The training was conducted using the Adam optimizer; a batch size is set to 16, and an initial learning rate of 0.001 is considered during training sessions. Given the classification task’s categorical nature, the categorical cross-entropy loss function is selected to reduce the disparity between predicted and actual class labels. The training process is iteratively executed for 100 epochs to refine the model’s final performance.

5 Results and Discussion

5.1 Detector

During the inferencing, all the test video sequences are fed to the detector in RGB format. The detector outputs the probability of exhibiting punches throughout each video as time series data. This time series data is passed through the truncated detector with threshold value *lambda*, followed by the Savitzky-Golay filter. The zero-crossing peak detection technique is applied to find the starting frame numbers of the detected punches. The detection accuracy is calculated by the formula given by the Eq. 5. The equation states that if any frame between the starting and ending frames of the detected punch lies between the starting and ending frames of ground truth punches, in a causal sense, it would taken as positive and vice-versa. The number of detected punches varies with threshold value λ , so average accuracy is calculated over different values of λ s, i.e., 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, and 0.9. The mean accuracy over all the test video subjects is found to be 96.09%.

$$Accuracy = \frac{\sum_{\lambda \in Thr} \frac{\sum_{P_i \in detP} I(P_i \cap P_j \geq 0, P_j \in L \cap \lambda)}{|L|}}{|Thr|} \quad (5)$$

The limitation of the detector is the threshold value, which is adjustable but depends on the view angle and subject. It is applied to avoid false detection and introduce high-frequency components for better peak detection.

5.2 Classifier

The clips containing the punches as detected by the detector in the first stage, are presented to the classifier for final classification. The punch clips are classified into six basic punches: Cross, Jab, Lead Hook, Lead Uppercut, Rear Hook, and Rear Uppercut. The 2D poses from the punch clips are extracted using AlphaPose [5]. 12 important joints out of 17 joints that are necessary for determining the type of punches are selected for final classification. The final performance of the 3D-CNN-based classifier is given in Table 1. For S11, as mentioned in the section Dataset Description, both the detection and classification models are

Table 1. Detection and Classification performance of the proposed models over various subjects.

Metric	S11	S21	S31	S41
Subject	S11	S21	S31	S41
Length	15.40	11.26	5.03	11.45
Detection Accuracy (%)	100	90.65	100	97.61
Total Punches	116/125	74/81	29/34	134/160
Cross	23/23	16/17	6/6	51/58
Jab	23/24	17/17	8/8	34/38
Lead Hook	16/21	24/24	2/3	20/23
Lead Uppercut	19/19	4/8	6/6	9/16
Rear Hook	8/10	13/15	3/7	9/10
Rear Uppercut	27/28	0/0	4/4	11/15
Classification Accuracy (%)	92.79	91.36	85.29	83.75

trained using the clips extracted from S11. The results corresponding to S11 are presented using a 70–30 split, where 70% are used for training, and the rest 30% are used for testing. The Precision, Recall, and F1 score performance metrics are given in Table 2.

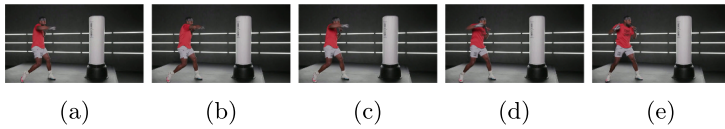


Fig. 6. Rear Hook Punch of S31 misclassified as Cross.

For S21, the model achieved an accuracy of 91.35%. In the case of S31, the proposed 3D-CNN achieved an accuracy of 85.29%. As shown in Table 2, the model achieved 100% precision, recall, and f1-score corresponding to Lead and Rear Uppercut. In the case of S31, 29 punches out of 34 detected punches are classified accurately. All Cross, Jab, Lead, and Rear Uppercut punches are correctly classified. As shown in Table 2, the precision, recall, and F1 score corresponding to class Rear Hook is comparatively poor. This is because most of the rear punches are misclassified as cross. Few frames from one of the rear hook punch mis-classified as cross is shown in Fig. 6. And finally, in case of S41, the model accurately classified 134 punches out of 160 punches detected. The most misclassification was reported in the Lead and Rear Uppercut category.

Since this study of fine-grained action recognition is performed on a novel YouTube video dataset (manually labelled), there is no state-of-the-art methodology that exist to be compared against the proposed work.

Table 2. Subject-wise performance metrics.

Subject	Performance Metrics	Types of punches					
		Cross	Jab	Lead Hook	Lead Uppercut	Rear Hook	Rear Uppercut
S11	Precision	0.82	0.96	0.93	1.00	1.00	1.00
	Recall	1.00	0.96	0.96	1.00	0.73	1.00
	F1 Score	0.90	0.96	0.95	1.00	0.84	1.00
S21	Precision	0.94	0.71	0.96	1.00	1.00	-
	Recall	0.94	1.00	1.00	0.67	0.50	-
	F1 Score	0.94	0.83	0.98	0.80	0.67	-
S31	Precision	0.60	0.89	1.00	1.00	1.00	1.00
	Recall	1.00	1.00	0.67	1.00	0.43	1.00
	F1 Score	0.75	0.94	0.80	1.00	0.60	1.00
S41	Precision	0.91	0.97	0.83	1.00	0.38	0.92
	Recall	0.88	0.89	0.87	0.56	0.90	0.73
	F1 Score	0.89	0.93	0.85	0.72	0.58	0.81

6 Conclusion

This paper presents a novel hierarchical framework tailored for robust sports activity recognition task. The proposed approach uses a fusion of regression-based detection employing a Dual Stream Spatio-Temporal Transformer followed by a fine-grained classification network powered by a 3D Convolutional Neural Network (CNN) for punch recognition from unedited boxing videos. Through rigorous experimentation on YouTube footage, the methodology achieves remarkable mean detection and classification accuracies of 96.09% and 88.25%, respectively, underscoring its robustness across diverse real-world scenarios. Beyond its immediate applications in sports performance analysis and athlete training, the research paves the way for broader implications, including real-time event monitoring and enhanced decision-making training for athletes. One of the major drawbacks of the proposed approach is that both the detector and the classifier are trained and tested using videos of frontal viewpoints only. In future, we aim to mitigate the drawback in case of boxing and make the pipeline view-invariant.

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